Mixed models:

Genetic relationships
Sire model
Individual (Animal) model

BLUP - reminder

- The MME provide a framework to obtain
  - BLUE of estimable k'b (fixed effects)
  - BLUP of random effects
- The properties of BLUP are more or less incorporated in the name:
  - Best ➔ maximises the correlation between the true effect (u), which is unknown, and the predicted genetic effect (û) or minimises prediction error variance (PEV) (var(u - û))
  - Linear ➔ predictors are linear functions of observations (data)
  - Unbiased ➔ estimates of a random effect (e.g. genetic value/merit of an individual) are unbiased (E(u) = û)); accounts for all other sources of systematic variation; also accounts for selection if all data on which selection has been based is included in the analysis
  - Prediction ➔ involves prediction of true effect

Numerator relationship (pedigree) matrix

- In lecture 1, we estimated the fixed effect solutions and predicted genetic merit of 3 sires (fathers) of daughters with phenotypes
- Assumed that the sires were unrelated
- In population studies, individuals tend to be related, and the genetic relationships are incorporated in the analyses
- The genetic covariance among individuals comprises three components:
  - additive genetic
  - dominance
  - epistatic
- Emphasis on the <u>additive</u> genetic component (why??)
- Use of additive genetic relationship matrix usually <u>increases the accuracy</u> of predictions and should help <u>account for previous selection</u> decisions if all pedigree (genealogical) data are utilised

Numerator relationship matrix

- The numerator relationship matrix (A) describes the additive genetic relationship among individuals

- The additive genetic relationship between individuals i and j is twice the probability of two genes taken at random from i and j being identical by descent
  - Common ancestor
  - Twice the co-ancestry or the coefficient of kinship

- The matrix A is symmetric and its
  - diagonal element for individual *i* (a_{ii}) is equal to 1 + F_i, with F_i is the inbreeding coefficient
  - off-diagonal element, a_{ij}, equals the numerator of the coefficient of relationship of individuals i and j

- When multiplied with genetic variance (σ^2_u) ➔ (co)variance of genetic values of individuals
  - $\mathrm{Var}(u_i) = a_{ii}\sigma^2_u = (1 + F_i)\sigma^2_u$
  - $\mathrm{Cov}(u_i, u_j) = a_{ij}\sigma^2_u$

Recursive method for computing A-matrix

Pedigree are coded 1 to *n* and ordered such that parents precede their progeny

If both parents (*s* and *d*) of individual *i* are known

$a_{ji} = a_{ij} = 0.5(a_{js} + a_{jd})$ $\quad j = 1$ to i-1

$a_{ii} = 1 + 0.5(a_{sd})$

If only one parent *s/d* is known and assumed unrelated to the mate

$a_{ji} = a_{ij} = 0.5(a_{js/d})$ $\quad j = 1$ to i-1

$a_{ii} = 1$

If both parents are unknown, they are assumed unrelated

$a_{ji} = a_{ij} = 0$ $\quad j = 1$ to i-1

$a_{ii} = 1$

Example pedigree

Individual	Parents: Paternal (Sire)	Parents: Maternal (Dam)
3	1	2
4	1	0
5	4	3
6	5	2

$a_{11} = 1$

$a_{22} = 1$

$a_{12} = 0 = a_{21}$

$a_{33} = 1 + 0.5(a_{12}) = 1$

$a_{13} = 0.5(a_{11}+a_{12}) = 0.5(1.0 + 0) = 0.5 = a_{31}$

$a_{23} = 0.5(a_{21}+a_{22}) = 0.5(0 + 1.0) = 0.5 = a_{32}$

$a_{34} = 0.5(a_{31}) = 0.5(0.5) = 0.25 = a_{43}$

$a_{35} = 0.5(a_{34} + a_{33}) = 0.5(0.25 + 1) = 0.625 = a_{53}$

$a_{55} = 1 + 0.5(a_{34}) = 1 + 0.5(0.25) = 1.125$

$a_{36} = 0.5(a_{35} + a_{32}) = 0.5(0.625 + 0.5) = 0.5625 = a_{63}$

Recursive method for computing A-matrix

Individual	Parents: Paternal (Sire)	Parents: Maternal (Dam)
3	1	2
4	1	0
5	4	3
6	5	2

	1	2	3	4	5	6
1	1.00	0.00	0.50	0.50	0.50	0.25
2	0.00	1.00	0.50	0.00	0.25	0.625
3	0.50	0.50	1.00	0.25	0.625	0.563
4	0.50	0.00	0.25	1.00	0.625	0.313
5	0.50	0.25	0.625	0.625	1.125	0.688
6	0.25	0.625	0.563	0.313	0.688	

a_{66} =

Recursive method for computing A-matrix

Individual	Parents: Paternal (Sire)	Parents: Maternal (Dam)
3	1	2
4	1	0
5	4	3
6	5	2

	1	2	3	4	5	6
1	1.00	0.00	0.50	0.50	0.50	0.25
2	0.00	1.00	0.50	0.00	0.25	0.625
3	0.50	0.50	1.00	0.25	0.625	0.563
4	0.50	0.00	0.25	1.00	0.625	0.313
5	0.50	0.25	0.625	0.625	1.125	0.688
6	0.25	0.625	0.563	0.313	0.688	

$a_{66} = 1 + 0.5(a_{52}) = 1 + 0.5(0.25) = 1.125$

From the above calculation the inbreeding coefficient for individuals 5 and 6 is 0.125

Decomposing the A-matrix

- A-matrix may be too large to handle
- A = T'DT
  - T is a lower triangular matrix
  - non-zero elements of T (t_{ij}) are the coefficients of relationship between individuals *i* and *j*
- T can be easily computed as per following rules:
  - For the ith individual: $t_{ii} = 1$
  - If both parents of i are known: $t_{ij} = 0.5(t_{sj} + t_{dj})$
  - If only one parent of i is known: $t_{ij} = 0.5(t_{sj})$
  - If neither parent of i is known: $t_{ij} = 0$

Matrix T for example pedigree

The matrix T is:

	1	2	3	4	5	6
1	1.0	0.0	0.0	0.0	0.0	0.0
2	0.0	1.0	0.0	0.0	0.0	0.0
3	0.5	0.5	1.0	0.0	0.0	0.0
4	0.5	0.0	0.0	1.0	0.0	0.0
5	0.5	0.25	0.5	0.5	1.0	0.0
6	0.25	0.625	0.25	0.25	0.5	1.0

Individual	Paternal (Sire)	Maternal (Dam)
3	1	2
4	1	0
5	4	3
6	5	2

Decomposing the A-matrix

D is a diagonal matrix of the co-variances for Mendelian sampling.

Mendelian sampling (*m*) for an individual *i:*

$$m_i = u_i - 0.5(u_s + u_d)$$

D for individual i is calculated as follows:

- both parents known: $d_{ii} = 0.5 - 0.25(F_s + F_d)$
- only one parent known: $d_{ii} = 0.75 - 0.25(F_s)$
- no parent known: $d_{ii} = 1$

- $F_i = 0.5a_{sd}$

Matrix D for example pedigree

- D = diag(1.0, 1.0, 0.5, 0.75, 0.5, 0.469)
- Inbred individuals: 5 and 6
  - 5 is also a parent
- Individual 4:
  - $d_{44} = 0.75 - 0 = 0.75$
  - Because only one parent is known and non-inbred
- Individual 6:
  - $d_{66} = 0.5 – [0.25(0.125 + 0)] = 0.469$
  - because both parents are known and the sire (father) has inbreeding coefficient of 0.125

Individual	Paternal (Sire)	Maternal (Dam)
3	1	2
4	1	0
5	4	3
6	5	2

A-inverse (A^{-1})

- We can reconstruct A from its components
  - A = T'DT
- But we are really interested in A-inverse
  - Remember MME??

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

Deriving rules for A-inverse

Individual	Paternal (Sire)	Maternal (Dam)
3	1	2
4	1	0
5	4	3
6	5	2

$A^{-1} = (T^{-1})'D^{-1}T^{-1}$

$T^{-1} = (I - M)$, where M is a matrix that traces genes from parents to offspring

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.5 & 0.5 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.5 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.5 & 0.5 & 0.0 & 0.0 \\ 0.0 & 0.5 & 0.0 & 0.0 & 0.5 & 0.0 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ -0.5 & -0.5 & 1.0 & 0.0 & 0.0 & 0.0 \\ -0.5 & 0.0 & 0.0 & 1.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & -0.5 & -0.5 & 1.0 & 0.0 \\ 0.0 & -0.5 & 0.0 & 0.0 & -0.5 & 1.0 \end{bmatrix}$$

- D^{-1} = diag(1,1,2,1.333,2,2.133)

Deriving rules for A-inverse

Therefore $A^{-1} = (T^{-1})'D^{-1}T^{-1}$ is:

	1	2	3	4	5	6
1	1.83	0.50	-1.00	-0.67	0.00	0.00
2	0.50	2.03	-1.00	0.00	0.53	-1.07
3	-1.00	-1.00	2.50	0.50	-1.00	0.00
4	-0.67	0.00	0.50	1.83	-1.00	0.00
5	0.00	0.53	-1.00	-1.00	2.53	-1.07
6	0.00	-1.07	0.00	0.00	-1.07	2.13

Direct rules for the A-inverse (ignoring inbreeding)

$A^{-1} = 0$

If both parents of the ith individual are known (s, d), add

2	to the (i,i) element
-1	to the (s,i), (i,s), (d,i) and (i,d) elements
0.5	to the (s,s), (s,d), (d,s) and (d,d) elements

If only one parent of the ith individual is known (s), add

1.333	to the (i,i) element
-0.667	to the (s,i) and (i,s) elements
0.333	to the (s,s) element

If neither parent of the ith individual is known, add

1	to the (i,i) element

Genetic analysis

- Uses pedigree data (A, A^{-1})
- Different software used to build and handle these matrices based on the different approaches and rules
- Assessment of genetic value/merit of individuals
- Agriculture/forestry
  - Estimated Breeding Value (EBV)
  - Predicted Transmitting Ability (1/2 EBV)

Sire model

- Using of the performance of progeny to evaluate the genetic merit of their sires
  - Through Artificial Insemination (livestock), sires may have hundreds, thousands or tens of thousands of progeny
- For example, the genetic evaluation (assessment of genetic merit) of dairy bulls can be calculated based on performance records of their daughters

Sire model

MME with the relationship matrix incorporated are

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

with $\lambda = \sigma^2_e / \sigma^2_s = (4-h^2)/h^2$

Where:

h^2 = var(u)/var(P) = heritability of the trait

var(P) = $\sigma^2_s + \sigma^2_e$

var(u) = $4 * \sigma^2_s$

Example: Sire model

Same data as in Lecture 1 , a sire model with A-matrix among the 3 sires incorporated.
Assume that $\lambda = \sigma^2_e/\sigma^2_s$ = 100/200 =0.50 and data as follows:

Cow	Herd	Season	Sire	Milk Yield (kg/d)
8	1	2	5	36.2
9	1	1	6	25.8
10	1	2	7	31.5
11	1	1	5	42.0
12	1	2	6	12.3
13	2	2	7	28.5
14	2	1	5	10.6
15	2	2	6	23.4
16	2	2	7	22.4
17	2	1	5	14.8

Sire	Sire of Sire	Dam of sire
5	1	0
6	2	3
7	2	4

Sires (previously 1, 2, 3) are recoded as 5, 6, 7, respectively, and cows (previously 1 to 10) are recoded from 8 to 17; ancestors of these sires will be coded from 1 to 4.

Example : Sire model

Sire	Sire of Sire	Dam of sire
5	1	0
6	2	3
7	2	4

Using the rules outlined earlier, the A^{-1} for the above pedigree is

$$A^{-1} = \begin{pmatrix} 1.333 & & & & & & \\ 0 & 2 & & & & & \\ 0 & 0.5 & 1.5 & & \text{symmetric} & & \\ 0 & 0.5 & 0 & 1.5 & & & \\ -0.667 & 0 & 0 & 0 & 1.333 & & \\ 0 & -1 & -1 & 0 & 0 & 2 & \\ 0 & -1 & 0 & -1 & 0 & 0 & 2 \end{pmatrix}$$

Example: Sire Model

- The matrix X is as defined in previous lecture
- Z is now set up to include the 4 ancestors in the pedigree plus the 3 sires (genetic effect)

$$Z' = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{pmatrix} \text{ and } Z'Z = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 \end{pmatrix}$$

- For λ =0.50:

$$Z'Z + A^{-1}\lambda = \begin{pmatrix} 0.667 & 0 & 0 & 0 & -0.333 & 0 & 0 \\ 0 & 1.00 & 0.25 & 0.25 & 0 & -0.5 & -0.5 \\ 0 & 0.25 & 0.75 & 0 & 0 & -0.5 & 0 \\ 0 & 0.25 & 0 & 0.75 & 0 & 0 & -0.5 \\ -0.333 & 0 & 0 & 0 & 4.667 & 0 & 0 \\ 0 & -0.5 & -0.5 & 0 & 0 & 4 & 0 \\ 0 & -0.5 & 0 & -0.5 & 0 & 0 & 4 \end{pmatrix}$$

Example : Sire Model

- **The MME are:**

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z+A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

$$\begin{pmatrix} 5 & & & & & & & & & & & \\ 0 & 5 & & & & & & \text{Symmetrical} & & & & \\ 2 & 2 & 4 & & & & & & & & & \\ 3 & 3 & 0 & 6 & & & & & & & & \\ 0 & 0 & 0 & 0 & 0.667 & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 1.00 & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0.75 & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0 & 0.75 & & & & \\ 2 & 2 & 3 & 1 & -0.333 & 0 & 0 & 0 & 4.667 & & & \\ 2 & 1 & 1 & 2 & 0 & -0.5 & -0.5 & 0 & 0 & 4.0 & & \\ 1 & 2 & 0 & 3 & 0 & -0.5 & 0 & -0.5 & 0 & 0 & 4.0 & \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \end{pmatrix} = \begin{pmatrix} 147.8 \\ 99.7 \\ 93.2 \\ 154.3 \\ 0 \\ 0 \\ 0 \\ 0 \\ 103.6 \\ 61.5 \\ 82.4 \end{pmatrix}$$

Example: Sire Model

		no pedigree (Lecture 1)
Herd		
1	11.239	11.330
2	0.000	0.000
Season		
1	0.000	17.510
2	2.465	19.817
Sire (random effect)		
1	0.915	
2	-0.915	
3	-3.155	
4	2.240	
5	1.830	1.909
6	-5.190	-5.230
7	2.903	3.320

Consistent results are generally expected

Not the same

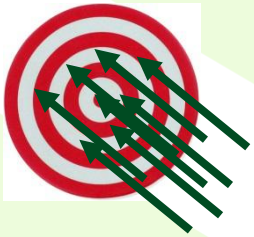
Including genetic relationships (A^{-1}):

- **Increases accuracy (B in BLUP)**
- **Underpins unbiasedness (U in BLUP)**

Unbiasedness vs. accuracy of prediction

Perfect prediction (on target, 100% accurate)
Ideal, but often unrealistic in real life

Unbiased prediction
High accuracy



Unbiased prediction
Low accuracy

High accuracy
Biased prediction

Accuracy and prediction error variance

- **The accuracy (r) of predictions is the correlation between true (unknown) and estimated random genetic effects.**

- ***In genetic evaluations (e.g. livestock, plants, trees), the accuracy is usually expressed in terms of reliability, which is r^2.***

- **Calculation for r or r^2 requires the diagonal elements of the inverse of the MME.**

Accuracy and prediction error variance

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

Coefficient Matrix (C)

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}$$

Accuracy and prediction error variance

- If the coefficient matrix of the MME is C:

$$\mathbf{C} = \begin{pmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{pmatrix} \text{ and a generalised inverse } \mathbf{C}^{-1} = \begin{pmatrix} \mathbf{C}^{11} & \mathbf{C}^{12} \\ \mathbf{C}^{21} & \mathbf{C}^{22} \end{pmatrix}$$

- Prediction error variance (PEV) = var(u-û) = $C^{22}\sigma^2_e$
- PEV could be regarded as the fraction of additive genetic variance not accounted for by the prediction. Therefore:
  - PEV = $C^{22}\sigma^2_e = (1 - r^2)\sigma^2_u$ or $(1 - r^2)\sigma^2_s$ for a sire model
  - with r^2 = squared correlation between the true and estimated breeding values
    - r and r^2 ➔ range from 0 to 1 (0-100%)

Accuracy and prediction error variance

- Thus for individual i:
  - $c_i\sigma^2_e = (1 - r^2)\sigma^2_s$
    - where c_i is the ith diagonal element of C^{22}
  - $c_i\ \sigma^2_e/\sigma^2_s = 1 - r^2$
  - $r^2 = 1 - c_i\lambda$

- Reliabilities for the 3 sires with data are 0.55, 0.57 and 0.54, respectively.

- Software such as ASReml calculates the standard error of prediction from which the accuracy and reliability may be derived (practical tutorial)

Individual (Animal) model

- In the previous example, the genetic merit of only the sires of individual cows were predicted.
  - Main advantage: number of equations are reduced since only sires are evaluated compared to all individuals in the data set.
  - Disadvantages: genetic merit of the mate (dam of progeny) is not accounted for; this may result in bias in the genetic analysis if there is preferential mating.

- A genetic analysis that predicts the genetic merit of all individuals in the dataset would overcome this problem. This is achieved with an individual (animal) model.

- The same MME are used to evaluate all individuals with records and all relatives in the pedigree. However, the λ term with an animal model changes to: $\lambda = \sigma^2_e/\sigma^2_u = (1-h^2)/h^2$

Individual (Animal) model

Individual	Pen	Sire	Dam	WG (kg)
4	1	1	0	4.5
5	2	3	2	2.9
6	2	1	2	3.9
7	1	4	5	3.5
8	1	3	6	5.0

Assume: $\sigma^2_u = 20$ and $\sigma^2_e = 40 \rightarrow \sigma^2_p = 60$

$\lambda = \sigma^2_e / \sigma^2_u = 40/20 = 2$

Also: $\lambda = (1-h^2)/h^2$

Where:

$h^2 = var(u)/var(P)$ = heritability of the trait (here $h^2 = 0.33$)

$var(P) = \sigma^2_p = \sigma^2_u + \sigma^2_e$

$var(u) = \sigma^2_u$ (*remember: in sire model* $var(u) = 4*\sigma^2_s$)

Incidence matrices

- **Observations vector** $y' = (4.5 \;\; 2.9 \;\; 3.9 \;\; 3.5 \;\; 5.0)$

- **Fixed effects (2 pens)** $X' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$

- **Random effects (8 individuals)**

$$Z = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad X'Z = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Matrices for the MME

$$\mathbf{X'y} = \begin{pmatrix} 13.0 \\ 6.8 \end{pmatrix}$$

$$\text{Tranpose of } \mathbf{Z'y} = \begin{pmatrix} 0 & 0 & 0 & 4.5 & 2.9 & 3.9 & 3.5 & 5.0 \end{pmatrix}$$

$$\mathbf{Z'Z} = diag(0,0,0,1,1,1,1,1)$$

The A^{-1} for the example data

$$A^{-1} = \begin{bmatrix} 1.833 & 0.500 & 0.000 & -0.667 & 0.000 & -1.000 & 0.000 & 0.000 \\ 0.500 & 2.000 & 0.500 & 0.000 & -1.000 & -1.000 & 0.000 & 0.000 \\ 0.000 & 0.500 & 2.000 & 0.000 & -1.000 & 0.500 & 0.000 & -1.000 \\ -0.667 & 0.000 & 0.000 & 1.833 & 0.500 & 0.000 & -1.000 & 0.000 \\ 0.000 & -1.000 & -1.000 & 0.500 & 2.500 & 0.000 & -1.000 & 0.000 \\ -1.000 & -1.000 & 0.500 & 0.000 & 0.000 & 2.500 & 0.000 & -1.000 \\ 0.000 & 0.000 & 0.000 & -1.000 & -1.000 & 0.000 & 2.000 & 0.000 \\ 0.000 & 0.000 & -1.000 & 0.000 & 0.000 & -1.000 & 0.000 & 2.000 \end{bmatrix}$$

Solutions to the MME

Pens		
1	4.358	BLUE
2	3.404	
Individuals		
1	0.098	û: Genetic Values (EBVs) BLUP
2	-0.019	
3	-0.041	
4	-0.009	
5	-0.186	
6	0.177	
7	-0.249	
8	0.183	

Understanding genetic/breeding values (EBV)

- EBV of an individual is made of contributions from:
  - Parents (PA)
  - Own record (YD)
  - Progeny (PC)

- $EBV = n_1PA + n_2YD + n_3PC$
  - Where $n_1 + n_2 + n_3 = 1$
  - Numerator
    - $n_1 = 2\lambda$, $4/3\lambda$ or 1λ if both, one or no parents are known
    - n_2 = number of records
    - $n_3 = \frac{1}{2}\lambda$ or $\frac{1}{3}\lambda$ if mate is known or unknown
  - Denominator
    - sum of numerators for n_1, n_2, n_3
  - PA = average EBV (û) of parents
  - YD = record of individual adjusted for fixed effects
  - PC = progeny EBV (û) adjusted for other parent's EBV (û)

EBV of individual 5: EBV (û)=-0.186)

u	EBV(û)	b	sol.	s	EBV(û)	d	EBV(û)	y
5		2	(3.404)	3	(-0.041)	2	(-0.019)	2.9
7	(-0.249)	1		4	(-0.009)	5		3.5

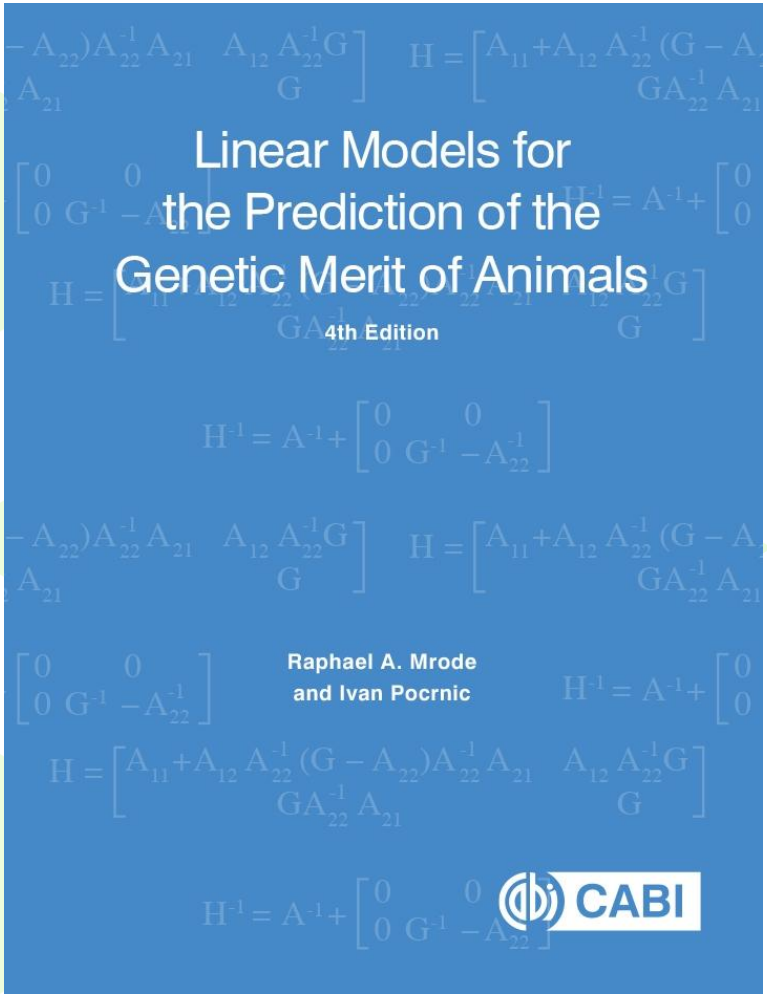
Numerators

$n_1 = 2\lambda = 4$
$n_2 = 1$ ➔ Denominator = 6 ➔ $n_1 = 4/6$, $n_2 = 1/6$, $n_3 = 1/6$
$n_3 = ½\lambda = 1$

PA = 4/6*[((-0.041)+(-0.019))/2] = 0.667*(-0.030) = -0.020
YD = 1/6*(2.9-3.404) = 0.167*(-0.504) = -0.084
PC = 1/6*[(2*(-0.249))-(-009)]=0.167*(-0.489) = -0.082

-0.186

Useful reading



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